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FURTHER MATHEMATICS

9231/34

Paper 3 Further Mechanics

May/June 2025

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 m s^{-2} .

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.

- 1 A light spring of natural length a and modulus of elasticity $20mg$ is placed so that it stands vertically on a horizontal plane. The lower end of the spring is fixed to the plane. A particle of mass m is attached to the upper end of the spring.

The particle is pushed vertically downwards until the length of the spring is $\frac{3}{5}a$. The system is then released from rest.

Find the maximum extension of the spring in the subsequent motion.

[5]

At max extension, $v=0 \Rightarrow KE=0$

Initially:

$$EPE = \frac{\lambda x^2}{2l} = \frac{20mg \left(\frac{2}{5}a\right)^2}{2a} = \frac{8}{5}mga \text{ J}$$

$$GPE = \frac{3}{5}mga \text{ J}$$

Finally:

$$EPE = \frac{\lambda x^2}{2l} = 10mg \frac{x^2}{a} \text{ J}$$

$$GPE = mg(x+a)$$

$$\rightarrow mgx + mga + 10mg \frac{x^2}{a} = \frac{11}{5}mga$$

$$x + a + 10 \frac{x^2}{a} = \frac{11}{5}a$$

$$10x^2 + ax - \frac{6}{5}a^2 = 0$$

$$x = \frac{-a \pm \sqrt{a^2 - 4(10)\left(-\frac{6}{5}a^2\right)}}{20}$$

$$= \frac{-a \pm \sqrt{49a^2}}{20} = \frac{6a}{20} \text{ or } -\frac{8a}{20}$$

$$\downarrow \left(\frac{3a}{10} \right) \text{ or } -\frac{4a}{10}$$

$$\rightarrow x = \frac{3a}{10}$$

$$\text{as } x, GPE > 0$$



- 2 A particle P of mass m kg moves along a horizontal straight line against a resistive force of magnitude $2mv^3$ N, where v ms⁻¹ is the velocity of P at time t s. When $t = 0$, $v = 1$.

(a) Find an expression for v in terms of t .

[4]

$$m \frac{dv}{dt} = -2mv^3$$

sub (0, 1)

$$\frac{1}{4} = C$$

$$\frac{dv}{2v^3} = -dt$$

$$\rightarrow \frac{1}{4v^2} = t + \frac{1}{4}$$

$$\frac{1}{2} \int v^{-3} dv = - \int dt$$

$$\frac{1}{v^2} = 4t + 1$$

$$\frac{1}{2} \left(\frac{-1}{2v^2} \right) = -t + C$$

$$v = \sqrt{4t+1}$$

$$\frac{1}{4v^2} = t + C$$

(b) Find the displacement of P from its initial position when $t = 6$.

[3]

$$x = \int_0^6 (4t+1)^{-1/2} dt$$

$$= \left[(2\sqrt{4t+1}) \left(\frac{1}{4} \right) \right]_0^6$$

$$= \frac{1}{2} \sqrt{25} - \frac{1}{2} \sqrt{1}$$

$$= 2m$$



- 3 A particle P is projected with speed u at an angle α above the horizontal from a point O on a horizontal plane and moves freely under gravity. The horizontal and vertical displacements of P from O at any subsequent time t are denoted by x and y respectively.

(a) Derive the equation of the trajectory of P in the form

$$y = x \tan \alpha - \frac{gx^2}{2u^2} \sec^2 \alpha. \quad [4]$$

Horizontally:

$$x = (v)(t) = (u \cos \alpha)(t) = ut \cos \alpha$$

Vertically:

$$\hookrightarrow \boxed{t = \frac{x}{u \cos \alpha}} \quad (1)$$

$$y = ut + \frac{1}{2}at^2$$

$$= ut \sin \alpha - \frac{1}{2}gt^2$$

$\hookrightarrow$ Subst (1)

$$y = u \left(\frac{x}{u \cos \alpha} \right) (\sin \alpha) - \frac{1}{2}g \left(\frac{x}{u \cos \alpha} \right)^2$$

$$= x \tan \alpha - \frac{1}{2}g \frac{x^2}{u^2 \cos^2 \alpha}$$

$$\boxed{y = x \tan \alpha - \frac{gx^2}{2u^2} \sec^2 \alpha}$$





It is given that $u = 20\sqrt{2} \text{ ms}^{-1}$ and that the particle P passes through the point where $x = 64 \text{ m}$ and $y = 8 \text{ m}$.

(b) Find the possible values of $\tan \alpha$.

[3]

$$\text{Subs } (u, x, y) = (20\sqrt{2}, 64, 8):$$

$$y = x \tan \alpha - \frac{gx^2}{2u^2} \sec^2 \alpha$$

$$8 = 64 \tan \alpha - 5 \left(\frac{4096}{800} \right) (1 + \tan^2 \alpha)$$

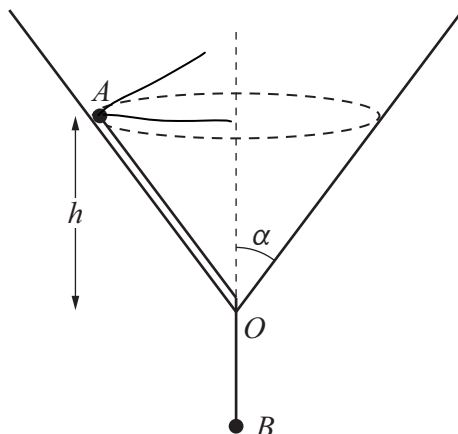
$$8 = 64 \tan \alpha - 25.6 - 25.6 \tan^2 \alpha$$

$$3.2 \tan^2 \alpha - 8 \tan \alpha + 4.2 = 0$$

$$0.2(4 \tan \alpha - 7)(4 \tan \alpha - 3) = 0$$

$$\hookrightarrow \boxed{\tan \alpha = 7/4 \text{ or } 3/4}$$





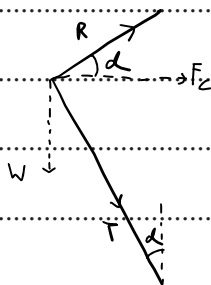
A hollow cone with a smooth inner surface is fixed with its vertex O downwards. The semi-vertical angle of the cone is α , where $\tan \alpha = \frac{3}{4}$. A light inextensible string has a particle A of mass m attached to one end and a particle B of mass m attached to the other end. The string passes through a small hole in the cone at O . Particle B hangs in equilibrium below O . Particle A is on the inner surface of the cone at a height h above the level of O and moves in horizontal circles with constant angular speed ω (see diagram).

Find ω in terms of g and h .

[6]

→ We know Tension, $T = mg$ from B.

Consider A:



$$F_c = mr\omega^2 = R \cos \alpha + T \sin \alpha \quad (1)$$

$$mg + T \cos \alpha = R \sin \alpha \quad (2)$$

We know:

$$\tan \alpha = \frac{3}{4}$$

$$\sin \alpha = \frac{3}{5}$$

$$\cos \alpha = \frac{4}{5}$$

$$r = h \tan \alpha = \frac{3h}{4}$$

$$T = mg$$

So:

$$(1) \quad \frac{3mh\omega^2}{4} = \frac{4}{5}R + \frac{3mg}{5}$$

$$(2) \quad mg = \frac{3R}{5} - \frac{4mg}{5}$$

$$\hookrightarrow \frac{9mg}{5} = \frac{3R}{5}$$

$$R = 3mg$$

$$\hookrightarrow \frac{3mh\omega^2}{4} = \frac{12mg}{5} + \frac{3mg}{5}$$

$$mh\omega^2 = 4mg$$

$$\omega^2 = \frac{4g}{h} \Rightarrow \omega = 2\sqrt{\frac{g}{h}}$$

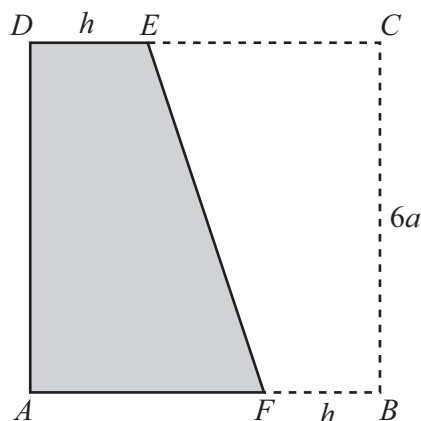




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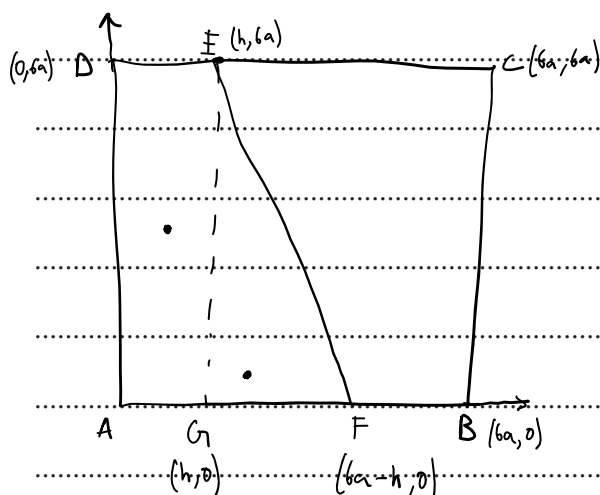




$ABCD$ is a uniform square lamina of side $6a$. Points E and F are on DC and AB respectively and are such that $DE = FB = h$. The quadrilateral $BCEF$ is removed from the square lamina (see diagram).

- (a) Show that the distance of the centre of mass of the resulting lamina $AFED$ from AD is $\frac{h^2 - 6ah + 36a^2}{18a}$ and find a corresponding expression for the distance of the centre of mass from AB . [5]

First consider:



Shape	Mass	COM
AGED	$6ah\rho$	$(\frac{h}{2}, 3a)$
GFEF	$(18a^2 - 6ah)\rho$	$(2a + \frac{h}{3}, 2a)$

About AD:

$$(\bar{x})(\rho)(18a^2 - 6ah + 6ah) = (\rho)(2a + \frac{h}{3})(18a^2 - 6ah) + (\rho)(\frac{h}{2})(6ah)$$

$$(\bar{x})(18a) = 36a^2 + 6ah - 12ah - 2h^2 + 3h^2$$

$$\bar{x} = \frac{h^2 - 6ah + 36a^2}{18a}$$

About AB:

$$(\bar{y})(\rho)(18a^2 - 6ah + 6ah) = \rho(2a)(18a^2 - 6ah) + \rho(3a)(6ah)$$

$$18\bar{y} = 36a - 12h + 18h$$

$$\bar{y} = \frac{6a + h}{3}$$

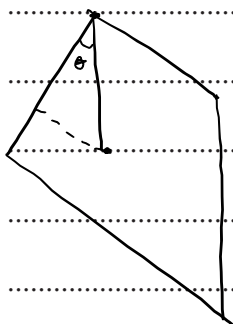




When the lamina $AFED$ is suspended from the point D , the edge DA makes an angle θ with the downward vertical, where $\tan \theta = \frac{7}{15}$.

(b) Find, in terms of a , the two possible values of h .

[3]



$$\tan \theta = \frac{7}{15}$$

$$\frac{7}{15} = \frac{\left(\frac{h^2 - 6ah + 36a^2}{18a} \right)}{\left(a - \left(\frac{6a + h}{3} \right) \right)}$$

$$\frac{7}{15} = \frac{h^2 - 6ah + 36a^2}{18a \left(4a - \frac{h}{3} \right)}$$

$$504a^2 - 42ah = 15h^2 - 90ah + 540a^2$$

$$0 = 36a^2 - 48ah + 15h^2$$

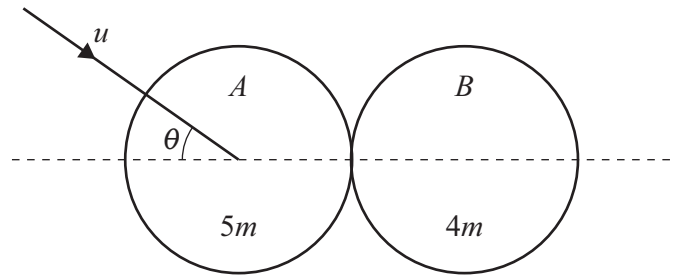
$$0 = 5h^2 - 16ah + 12a^2$$

$$0 = (5h - 6a)(h - 2a)$$

$$h = 2a \quad \text{or} \quad h = \frac{6a}{5}$$



6



Two uniform smooth spheres A and B of equal radii have masses $5m$ and $4m$ respectively. Sphere A is moving with speed u on a horizontal surface when it collides with sphere B which is at rest. Immediately before the collision, A 's direction of motion makes an angle θ with the line of centres (see diagram). The coefficient of restitution between the spheres is e .

- (a) Show that the speed of B after the collision is $\frac{5}{9}u(1+e)\cos\theta$. [3]

Com:

$$5mv\cos\theta = 5mv_A + 4mv_B$$

$$5v\cos\theta = 5v_A + 4v_B$$

Newton's:

$$\frac{v_B - v_A}{v\cos\theta} = e$$

$$v_A = v_B - ev\cos\theta$$

$$\hookrightarrow 5v\cos\theta = 5v_B - 5ev\cos\theta + 4v_B$$

$$\boxed{\frac{5v\cos\theta(1+e)}{9} = v_B}$$



After the collision the kinetic energy of A is equal to the kinetic energy of B .

- (b) Given that $\tan \theta = \frac{2}{3}$, find the value of e .

[6]

$$\text{Initially, } KE = \frac{1}{2}(5m)(v^2) = \frac{5}{2}mv^2$$

$$\tan \theta = \frac{2}{3}$$

$$\sin \theta = \frac{2}{\sqrt{13}}$$

$$\text{Finally: } v_B = \frac{5}{9}v \left(\frac{3}{\sqrt{13}} \right) (1+e) = \frac{5v}{3\sqrt{13}} (1+e)$$

$$\cos \theta = \frac{3}{\sqrt{13}}$$

$$v_{xA} = \frac{5}{9}v \left(\frac{3}{\sqrt{13}} \right) (1+e) - ev \left(\frac{3}{\sqrt{13}} \right)$$

$$= \frac{v}{\sqrt{13}} \left(\frac{5}{3} + \frac{5}{3}e - 3e \right)$$

$$= \frac{v}{\sqrt{13}} \left(\frac{5}{3} - \frac{4}{3}e \right)$$

$$v_{yA} = \frac{2v}{\sqrt{13}}$$

$$KE_B = \frac{1}{2}(4m) \left(\frac{5v}{3\sqrt{13}} \right)^2 (1+e)^2$$

$$= \frac{50}{117}mv^2 (e^2 + 2e + 1)$$

$$KE_A = \frac{1}{2}(5m) \left(\frac{4v^2}{13} + \frac{v^2}{13} \left(\frac{5}{3} - \frac{4}{3}e \right)^2 \right) = \frac{5}{26}mv^2 \left(4 + \left(\frac{5}{3} - \frac{4}{3}e \right)^2 \right)$$

$$\hookrightarrow KE_B = KE_A$$

$$\frac{50}{117}mv^2 (e^2 + 2e + 1) = \frac{5}{26}mv^2 \left(4 + \frac{25}{9} + \frac{16}{9}e^2 - \frac{40}{9}e \right)$$

$$\frac{20}{9} (e^2 + 2e + 1) = \frac{1}{9} (16e^2 - 40e + 61)$$

$$4e^2 + 80e - 41 = 0$$

$$4e^2 + 82e - 2e - 41 = 0$$

$$2e(2e + 41) - (2e + 41) = 0$$

$$e = \frac{1}{2} \text{ or } -\frac{41}{2}$$

$$\hookrightarrow e < 1$$

$$\hookrightarrow e = \frac{1}{2}$$





- 7 A particle P of mass m is attached to one end of a light inextensible string of length a . The other end of the string is attached to a fixed point O . The particle moves in complete vertical circles with centre O , with the string taut. When the string makes an angle θ with the downward vertical through O the speed of P is $\sqrt{4ag}$. The ratio of the greatest and least tensions in the string during the motion is $11 : 1$.

Find the value of $\cos \theta$.

[8]

Taking initial speed $= u$

$$T_{\max}: \theta = 0 \rightarrow T = \frac{mv^2}{a} - 2mg + 3mg(\cos 0) \\ = \frac{mv^2}{a} + mg$$

$$T_{\min}: \theta = 180 \rightarrow T = \frac{mv^2}{a} - 2mg + 3mg(\cos 180) \\ = \frac{mv^2}{a} - 5mg$$

$$\rightarrow 11 \left(\frac{mv^2}{a} - 5mg \right) = \frac{mv^2}{a} + mg$$

$$\frac{11v^2}{a} - 55g = \frac{v^2}{a} + g$$

$$\frac{10v^2}{a} = 56g$$

$$v = \sqrt{5.6ga}$$

$$@ \theta, v = \sqrt{4ag}$$

$$\hookrightarrow \frac{mv^2}{a} = \frac{mv^2}{a} - 2mg + 2mg \cos \theta$$

$$4g = 5.6g - 2g + 2g \cos \theta$$

$$0.4 = 2 \cos \theta$$

$$\cos \theta = \frac{1}{5}$$





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Additional Page

If you use the following lined page to complete the answer(s) to any question(s), the question number(s) must be clearly shown.

Lined area for writing answers, consisting of multiple horizontal dotted lines.







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